Import libraries

```
# Importing Libraries
# Basic Libraries
import numpy as np
import pandas as pd

# Visualization
import seaborn as sns
import matplotlib.pyplot as plt
from wordcloud import WordCloud

# ML Utilities
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder, StandardScaler

# Feature Extraction
from sklearn.feature_extraction.text import TfidfVectorizer

# Feature Selection
from sklearn.feature_selection import chi2

# ML Models
from sklearn.neighbors import KNeighborsClassifier
from sklearn.naive_bayes import MultinomialNB

# Metrics
from sklearn.metrics import accuracy_score, confusion_matrix
```

Load The DataSet

```
# Loading The Dataset
df = pd.read_csv('/content/IMDB Dataset.csv')
```

```
# Print The Shape Of Data
df.shape
```

```
(50000, 2)
```

```
# Print The Data
df
```

	review	sentiment
0	One of the other reviewers has mentioned that ...	positive
1	A wonderful little production. The...	positive
2	I thought this was a wonderful way to spend ti...	positive
3	Basically there's a family where a little boy ...	negative
4	Petter Mattei's "Love in the Time of Money" is...	positive
...	...	...
49995	I thought this movie did a down right good job...	positive
49996	Bad plot, bad dialogue, bad acting, idiotic di...	negative
49997	I am a Catholic taught in parochial elementary...	negative
49998	I'm going to have to disagree with the previou...	negative
49999	No one expects the Star Trek movies to be high...	negative

50000 rows × 2 columns

Next steps: Generate code with df | New interactive sheet

```
# Print The Dataset Information
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 50000 entries, 0 to 49999
Data columns (total 2 columns):
 #   Column     Non-Null Count  Dtype 
---  ------     --------------  ----- 
 0   review     50000 non-null  object
 1   sentiment  50000 non-null  object
dtypes: object(2)
memory usage: 781.4+ KB
```

Preprocess The Dataset

1. Missing Values

```
# Checking For Missing Values
df.isnull().sum()
```

	0
review	0
sentiment	0

dtype: int64

2. Duplicated Values

```
# Checking For Duplicated Values
print('Number Of Duplicated Values In The Dataset: ', df.duplicated().sum())
```

```
Number Of Duplicated Values In The Dataset:  418
```

```
# drop  Duplicated Values
df = df.drop_duplicates()
print('Number Of Duplicated Values In The Dataset after drop: ', df.duplicated().sum())
```

```
Number Of Duplicated Values In The Dataset after drop:  0
```

3. Class Imbalance

```
# Check class imbalance
print(df['sentiment'].value_counts())
print("\nPercentage:\n", df['sentiment'].value_counts(normalize=True) * 100)
```

```
sentiment
positive    24884
negative    24698
Name: count, dtype: int64

Percentage:
 sentiment
positive    50.187568
negative    49.812432
Name: proportion, dtype: float64
```

TF -IDF

```
vectorizer = TfidfVectorizer(max_features=3000, stop_words='english')
X_tfidf = vectorizer.fit_transform(df['review'])

df_tfidf = pd.DataFrame(X_tfidf.toarray(), columns=vectorizer.get_feature_names_out())
df_tfidf['sentiment'] = df['sentiment'].map({'positive':1, 'negative':0})

df_tfidf = df_tfidf.dropna()

X = df_tfidf.drop('sentiment', axis=1)
y = df_tfidf['sentiment']
```

Chi-Square Feature Importance

```
chi_scores, p_values = chi2(X, y)
```

```
chi2_results = pd.DataFrame({
    'feature': X.columns,
    'chi2_score': chi_scores,
    'p_value': p_values
})

chi2_results = chi2_results.sort_values(by='chi2_score', ascending=False)
chi2_results.head(20)
```

	feature	chi2_score	p_value
952	fairy	3.493275	0.061619
223	bad	3.356006	0.066960
1045	flynn	3.263206	0.070850
2897	waste	2.868435	0.090333
898	excellent	2.521037	0.112337
2974	worst	2.504976	0.113487
857	enjoyed	2.286419	0.130511
348	brooks	2.221555	0.136096
2231	romantic	2.105255	0.146794
1917	passion	2.089010	0.148362
2771	turkey	2.027960	0.154428
1896	paid	1.987753	0.158576
338	brilliant	1.919453	0.165917
2928	western	1.917130	0.166173
1306	hospital	1.822030	0.177072
1742	money	1.815215	0.177884
2682	themes	1.758497	0.184812
1616	low	1.701876	0.192043
1094	friendly	1.639103	0.200449
768	dollar	1.601754	0.205655

Next steps: Generate code with chi2_results | New interactive sheet

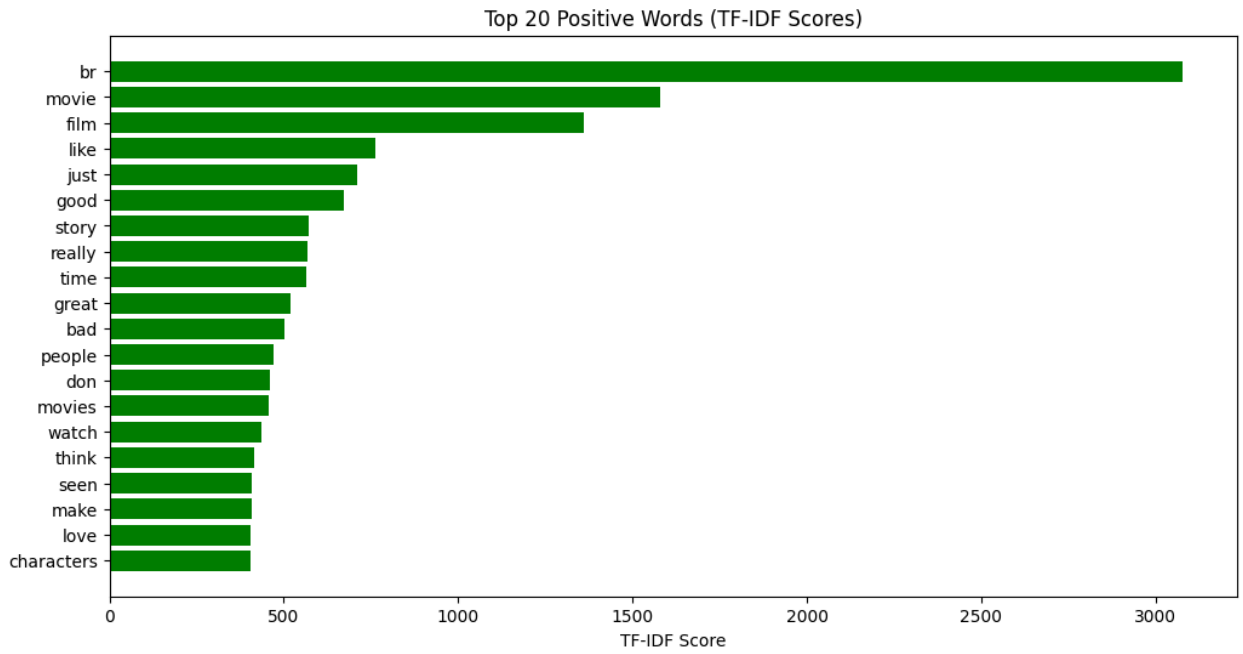
Top Positive vs Top Negative Words

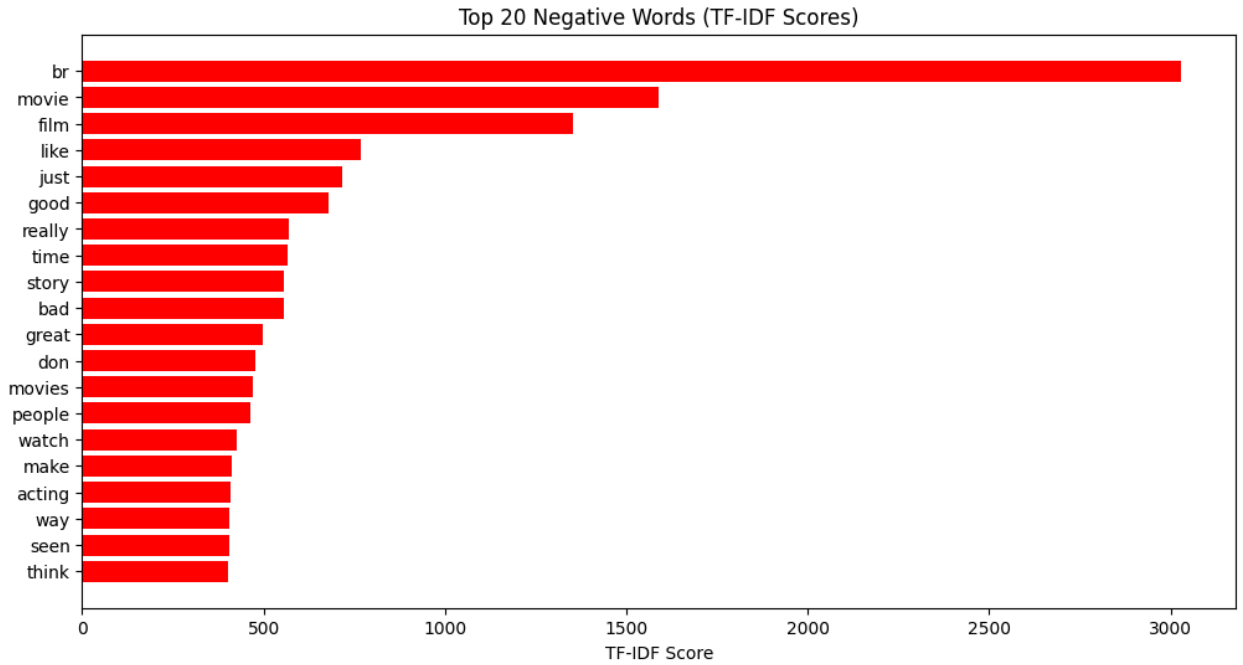
```
# Extracting TF-IDF words
feature_names = vectorizer.get_feature_names_out()

# Separate data by sentiment
pos_tfidf = df_tfidf[df_tfidf["sentiment"] == 1].drop("sentiment", axis=1)
neg_tfidf = df_tfidf[df_tfidf["sentiment"] == 0].drop("sentiment", axis=1)

# Total TF-IDF per word within each category
pos_scores = np.array(pos_tfidf.sum(axis=0)).flatten()
neg_scores = np.array(neg_tfidf.sum(axis=0)).flatten()

# Word order
top_pos_idx = pos_scores.argsort()[::-1][:20]   # Top 20 positive words
top_neg_idx = neg_scores.argsort()[::-1][:20]   # Top 20 negative words

top_positive_words = [(feature_names[i], pos_scores[i]) for i in top_pos_idx]
top_negative_words = [(feature_names[i], neg_scores[i]) for i in top_neg_idx]

print("Top Positive Words:")
print(top_positive_words)

print("-----------------------------------")

print("\nTop Negative Words:")
print(top_negative_words)
```

```
Top Positive Words:
[('br', np.float64(3078.507522805172)), ('movie', np.float64(1579.3008032414361)), ('film', np.float64(1359.8676682461842)),
```

```
------------------------------------

Top Negative Words:
[('br', np.float64(3029.2188962813593)), ('movie', np.float64(1587.2461197768196)), ('film', np.float64(1353.513351818457)),
```

```
# Convert to separate lists
pos_words = [w for w, s in top_positive_words]
pos_values = [s for w, s in top_positive_words]

neg_words = [w for w, s in top_negative_words]
neg_values = [s for w, s in top_negative_words]

plt.figure(figsize=(12, 6))
plt.barh(pos_words[::-1], pos_values[::-1], color='green')
plt.title("Top 20 Positive Words (TF-IDF Scores)")
plt.xlabel("TF-IDF Score")
plt.show()

plt.figure(figsize=(12, 6))
plt.barh(neg_words[::-1], neg_values[::-1], color='red')
plt.title("Top 20 Negative Words (TF-IDF Scores)")
plt.xlabel("TF-IDF Score")
plt.show()
```

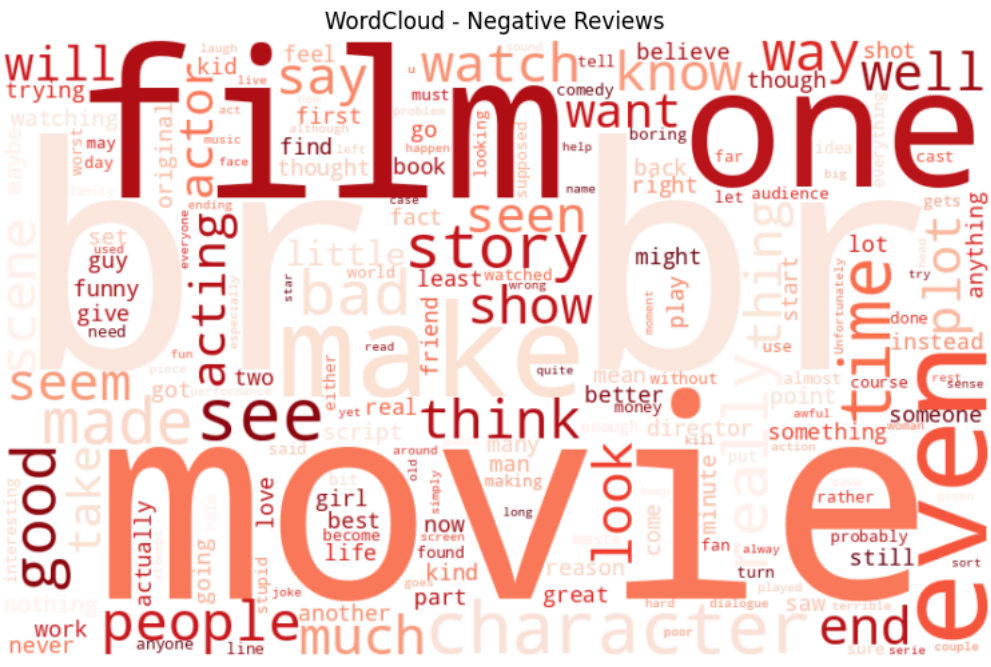





```
# Positive texts
pos_text = " ".join(df[df["sentiment"] == "positive"]["review"].astype(str))
```

```
# WordCloud Positive
wc_pos = WordCloud(width=800, height=500, background_color='white', colormap='Greens').generate(pos_text)
plt.figure(figsize=(10, 6))
plt.imshow(wc_pos, interpolation='bilinear')
plt.axis("off")
plt.title("WordCloud - Positive Reviews")
plt.show()
```


```
#  Negative texts
neg_text = " ".join(df[df["sentiment"] == "negative"]["review"].astype(str))

# WordCloud Negative
wc_neg = WordCloud(width=800, height=500, background_color='white', colormap='Reds').generate(neg_text)
plt.figure(figsize=(10, 6))
plt.imshow(wc_neg, interpolation='bilinear')
plt.axis("off")
plt.title("WordCloud - Negative Reviews")
plt.show()
```



Training the model

```
# X = texts
X = df['review']

# y = lebals
y = df['sentiment']

# Data breakdown: 70% training and 30% testing
X_train, X_test, y_train, y_test = train_test_split(
```

```
    X, y,
    test_size=0.3,        # 30%
    random_state=42,
    stratify=y            # Keeps the distribution balanced
)

print("Training samples:", len(X_train))
print("Testing samples:", len(X_test))
```

```
Training samples: 34707
Testing samples: 14875
```

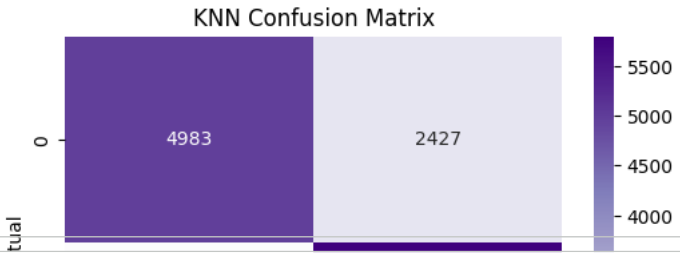
Testing the model

```
# X = Text Features (TF-IDF)
X = X_tfidf                      # From TfidfVectorizer

# y = Labels (positive / negative)
le = LabelEncoder()
y = le.fit_transform(df['sentiment'])   # It converts them to 0 or 1

# Data breakdown: 70% training - 30% testing
X_train, X_test, y_train, y_test = train_test_split(
    X, y,
    test_size=0.3,
    random_state=42,
    stratify=y        # Keeps the distribution balanced
)
```

```
# --------- KNN MODEL ---------
knn_model = KNeighborsClassifier(n_neighbors=5)

# Model training
knn_model.fit(X_train, y_train)

# Prediction based on training and testing
y_train_pred_knn = knn_model.predict(X_train)
y_test_pred_knn  = knn_model.predict(X_test)

# Accuracy calculation
train_acc_knn = accuracy_score(y_train, y_train_pred_knn)
test_acc_knn  = accuracy_score(y_test,  y_test_pred_knn)

print(f"KNN Train Accuracy : {train_acc_knn:.3%}")
print(f"KNN Test  Accuracy : {test_acc_knn:.3%}")
```

```
KNN Train Accuracy : 82.969%
KNN Test  Accuracy : 72.444%
```

```
# Confusion Matrix
cm_knn = confusion_matrix(y_test, y_test_pred_knn)

plt.figure(figsize=(6,4))
sns.heatmap(cm_knn, annot=True, fmt="d", cmap="Purples", cbar=True)
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.title("KNN Confusion Matrix")
plt.show()
```




```
# --------- Naive Bayes MODEL ---------
nb_model = MultinomialNB()

# Model training
nb_model.fit(X_train, y_train)

# Prediction based on training and testing
y_train_pred_nb = nb_model.predict(X_train)
y_test_pred_nb  = nb_model.predict(X_test)
```